The Role of Open Source Software in IoT

With more and more devices getting connected with each other, it's pretty clear that the Internet of Things (IoT) is going to play a major role in our lives in the days to come. In the not too distant future, smart automobiles and transportation, smart fans and household devices, and smart buildings with smart lighting will all be part of everyday life. Let's find out why open source software will provide the best platforms for the IoT.

All of us book a cab and order food through mobile apps today, as well as upload our status on social media and tweet our views about current happenings using our mobile phones. Now imagine your phone without the Internet connection – it's just a box that can be used only for audio calls and some basic operations. So, basically, the Internet is the one thing that gives value to a phone and makes it smart.

Similarly, other objects like TVs, ACs, fans, doors, watches, etc, can also be used smartly with the help of the Internet. This is what the Internet of Things (IoT) is all about. It connects our devices to the Internet and then to each other, using software to perform routine tasks in our daily lives.

Can a car talk to a house? Well, your car may soon tell your house that you are five miles away and instruct it to turn on the lights and the heating. So the core idea in IoT is to connect every single object to the Internet and also to one another.

Open source software (OSS) is used to perform many tasks. It is software with its source code accessible to everyone so that anyone can inspect, modify and enhance it according to their needs. Most people tend to choose open source software because they have more control over it—for instance, to change that part of the software that is of no use. Additionally, open source software is more secure and stable as anyone can fix, update and upgrade it, which happens more frequently than with proprietary software. Basically, any program starts with developers writing the code and then they set the rules of usage as the licence, which describes the extent to which contributors can change the code by the process of upstreaming.

How big is the IoT?

According to a study by Gartner, till last year there were more than 3.5 billion connected things in the form of smart cars, smoke detectors, locks, robots, music systems, health equipment, trains, wind turbines and more. This number will cross 25 billion by 2020, and that will be the start of what many call the fourth Industrial Revolution (the earlier three were driven by steam, then electricity and, most recently, wired computers).

Here, AI will play a major role as it will not only collect data from these devices but will also train 'things' to work according to a user's needs. So IoT will cover everything — from the automotive, mechanical and medical industries to hospitality, education and development. In the coming years, everything that you can think of will be on the Internet — your fan, watch, lights, bottles, clothes…everything. These things will have their own IP addresses, depending upon their memory and RAM. New protocols like CoAP will be in place for these tiny and constrained devices.

Challenges facing IoT

Security and vulnerability: Security is one of the most important pillars of the Internet and *the* most significant challenge facing the IoT. As the number of connected devices increases, the number of those that are poorly designed and offer opportunities to exploit security vulnerabilities also increases. This can expose user data to hackers by leaving data streams not properly protected and, in some cases, people's health and safety can also be put at risk. Although it's true that the IoT is more secure than the average Internet or LAN connection, it's not exactly the bulletproof shell that some users expect.